Requirements

Developer Accounts

Another type of accounts for developers is added, besides the Company and User accounts. The developer account does not have access to recommandations/status/chat pages but they have access to the developer page account. In the developer page, any developer can post a proposal. A proposal is a post that can be of 2 types:

Parameter proposal: parameter + value

A proposal of a parameter that serves a role in the compatibility scoring system is proposed (the mitigation factor, the weight for the skill distance, the weight for the resume-job similarity score , the weight for the user preference score or the weight of the promotion score) and its value.

Example: developer A can post: resume-job similarity score, weight 45%, meaning that that developer considers that the resume-job similarity score should have a weight of 45% in the compatibility score calculation.

The developer can select one of the 5 parameters mentioned earlier and add the value in a text box. For the parameters resembling weights, the system will give an error if the value is not a number between 0 and 100. For the parameter “mitigation factor” , the system will give an error if the value is not a number greater or equal than 1.

Keyword proposal

A proposal of a specific keyword that the developer considers to be important in the resume-job similarity score.

For example, developer A can post that AI is a relevant keyword, by writing “AI” into a text box after selecting the keyword option in the selection box.

Likes and dislikes

For any of the posts above, other developers can like/dislike them, by clicking the like or the dislike button below the post. The likes/dislikes will account for the compatibility scoring system that will change based on these developer proposals/reactions. A developer cannot like/dislike the same post multiple times (if a developer likes the post 2 times, the like will be deleted, same for dislikes). More details about how these actions influence the algorithm will be listed in the Compatibility Scoring System requirement.

Posting

A developer can create a new post by clicking a ‘New Post’ button. A pop-up will appear that contains a selection box with the options: “mitigation factor”, “weighted distance score weight”, “job-resume similarity score weight”, “preference score weight”, “promotion score weight”, “relevant keyword”, and a text box where you can add either the value for a selected parameter or a relevant, non-empty, all-lowercase keyword. In the pop-up there is a button named ‘post’, that if clicked either displays an error if the input is invalid, or posts the post.

Instructions

The matching algorithm will have a short explanation on the top of the page for the developers that are not familiar with it.

2) Compatibility Scoring System

The system should calculate a compatibility score between 0 and 100, where 0 means low compatibility and 100 means high, for every user, job pair. The score determines the relevance of a job to a user (and vice versa) and will be used as the core ranking system in the matchmaking logic. The score is computed using a weighted average between 4 distinct metrics:

Weighted Skill Distance

For each skill from the job’s desired skills, we take the matching skill from the user’s skills (if the user doesn’t have that skill recorded it will be defaulted to 0), then we compute this asymmetric difference logic: if the candidate is overqualified (user’s skill is higher than the job’s desired skill), then we take their difference and we divide it by a mitigation factor greater or equal than 1, otherwise we just take the difference. Then, the system shall compute the average of the asymmetric differences for every skill and subtract that from 100. If the job doesn’t have any desired skills or the final operation makes the result lower than 0, the final result is set to 0.

Resume-Job similarity search

The system shall normalize both the user’s resume and the job description by converting all text to lowercase and removing punctuation. The preprocessed text shall be converted into two unique sets of terms to ensure only distinct keywords are evaluated. The system shall determine the similarity coefficient by calculating the ratio of the scores of the intersection of these sets and their union. The score of a set is calculated by summing up the value for every keyword in the set. The value of a keyword is determined by the developers with the formula

*value(keyword) = 1 + 0.1 * sum(post_likes - post_dislikes)* for every post mentioning that specific keyword. The absolute value of *value(keyword)* is capped in the interval [0, 5] to avoid extreme or negative values.

If the union set is empty, this ratio will be defaulted to 0. The final result is then multiplied by 100.

User preference score

The system will compare the job's location with the user's preferred location and the job's employment type (internship, full-time, part-time, etc) with the user's preferred employment type. The final result is the percentage of these user's preferences that match the job's details.

Job Promotion Score

The system shall use the promotion level (number between 0 and 100) for the specific job as it is. If the promotion level is not defined or null, it will be defaulted to 0.

The system shall compute the final compatibility score using a weighted average of the 4 components.

The weights/mitigation factor are computed based on the values proposed by the developers in this manner:

value = sum(value proposed in a post max(0, post_likes - post_dislikes))*

If the value of a weight is not proposed in any post, it will be defaulted to 25%.
If the value of the mitigation factor is not proposed in any post, it will be defaulted to 2.

The weights may not sum up to 100% so they will be linearly scaled so that their sum equals exactly 100% in this manner:

*Weight = Weight * 100% / Sum(weights)*

2) Matchmaking Page: User

The Matchmaking Page is the core of the application from a user's perspective. It presents a stack of job posting cards, allowing users to express interest or disinterest in a position with a simple button click to apply or skip the job.

When the user opens the application, the home page displays a stacked deck of job cards. Each card shows a preview of the role, including the company name, job title, required skills, and a short description. The user can interact with the deck by tapping the action buttons below, or undoing their last action.

Before browsing, users can open a filter panel to narrow which job postings appear. Filters include job type, experience level, location, and required skills.

If no job postings are available under the active filters, a friendly empty state screen is shown, prompting the user to broaden their filters or refresh.

1. Card Stack

The home page displays a stacked deck of job cards sourced from the recommendation engine. Cards are shown one at a time, with the top card always being the next recommended posting. Jobs the user has already seen are not shown again if not enough time has passed

1.1 Display Card Stack

When the user opens the home page, the next recommended job is loaded and displayed as the top card of the stack. The card must display:

- Company name and logo
- Job title
- Location
- Job type
- Up to three required skills
- A description excerpt of no more than 150 characters

1.2 Like: Express Interest

When the user presses the like button, their interest in the role is recorded. The card is dismissed and the next card becomes the top card. If the action fails due to an error, an error message is shown and the card remains visible so the user can try again.

1.3 Dismiss: Skip Posting

When the user presses the dismiss button, the posting is skipped and will not reappear. The card is dismissed and the next card becomes the top card. If the action fails due to an error, an error message is shown and the card remains visible so the user can try again.

1.4 Action Buttons

Below the card stack, two action buttons are displayed at all times:

- Red button → skip the current posting
- Green button → apply to the current posting

Both buttons must be clearly visible and large enough to tap easily. The buttons are greyed out and inactive when no card is currently loaded.

1.5 Expand Card to Full Detail View

The user can tap anywhere on the visible card to expand it into a full detail view. This shows all the information from the card preview plus:

- Full job description
- All required skills and their minimum score thresholds
- Contact info

A Dismiss and a Like button at the bottom of the view work the same as the main buttons. A back button closes the view and returns the user to the card stack without triggering any action.

1.6 Undo Last Action

The user can undo their most recent application by pressing the Undo button, shown as a back-arrow icon below the card stack. Pressing it reverses the last action and returns the card to the top of the stack. Only one undo is available per session. The Undo button appears greyed out and inactive at the start of a session and after an undo has already been used.

2. Filters

Users can filter the job postings shown in the card stack before or during browsing.

2.1 Open and Configure Filter Panel

The user can open a filter panel by pressing the filter icon in the top-right corner of the home page. The panel slides in from the right and presents four options: job type as a multi-select field (full-time, part-time, internship, volunteer, remote, hybrid), experience level as a multi-select field (internship, entry level, mid-senior level, director, executive), location as a free-text input, and required skills as a multi-select dropdown.

2.2 Apply and Reset Filters

When the user presses Apply, the filter panel closes and the card stack refreshes to show only matching postings. Jobs the user has already interacted with are excluded regardless of filter changes. When the user presses Reset, all filters are cleared and pressing Apply again reloads the full unfiltered deck.

3. Empty State

3.1 Display Empty State Screen

When no further job postings are available under the active filters, the card stack is replaced by an empty state screen. The screen displays a descriptive message, a link to open the filter panel so the user can broaden their search, and a Refresh button to check for newly added postings.

4. Validation

The like and dismiss actions are only available when a card is currently loaded. Both buttons are greyed out and non-interactive when no card is present. If a network error occurs and the action cannot be completed, the card remains visible and an error message is shown so the user can retry.

5. Usability

The interactions must be intuitive and require no tutorial for first-time users. All core actions(liking, skipping, undoing, and filtering) must be reachable within two taps from the home screen. Touch targets for the like and dismiss buttons must be large enough to click with ease.

6. Security

The matchmaking page only shows data for the currently logged-in user. All actions are validated against the active session before being executed. Users must not be able to view or influence the activity or records of other users through the interface.

3) User Status Page

1. General Overview

The User Status Page is where the logged-in user can see all the jobs they have applied to, along with the current status of each application. The page only shows information the user cannot change anything from here, and they can only see their own applications.

The Skill Gap Analysis is a separate page that the user can open from the Status Page. It looks at the jobs the user was rejected from and compares the skills those jobs required with the skills the user has, showing which skills they are missing and which ones they need to improve.

2. User Status Page

2.1 View application status list

The user shall be able to reach the Status Page from the main navigation bar. When the page opens, the system loads all the applications the user has made and shows them as a scrollable list of cards, each one showing the job and company information. The page should be ready to use within 1 second of opening.

2.2 Status-based visual differentiation

Each job card shall show the current status of that application using a coloured badge with a text label. **Applied** shows a blue badge, meaning the company has not made a decision yet. **Accepted** shows a green badge, meaning the company accepted the user. **Rejected** shows a red badge, meaning the company rejected the user. The badge must always show both the colour and the text so it is clear even without colour. It must be visible directly on the card without needing to open it.

2.3 Job card content

Each card shall show the company name, the first 120 characters of the job description followed by "..." if it is longer, the date the user applied formatted as "Applied on DD MMM YYYY", the status badge, the compatibility score shown as a percentage (example: "74% match"), and the feedback (if it exists)

2.4 Filter by status

At the top of the list there shall be four filter buttons: **All**, **Applied**, **Accepted**, and **Rejected**. They must be visible without scrolling. When the user taps one, the list updates immediately to show only the cards matching that status. By default, **All** is selected. The list must update within 200 milliseconds of pressing a filter.

2.5 Expand job card to full detail view

The user can tap any card to open a full-screen view with more details about that job: the full job description, the list of required skills with the minimum score needed for each, the company name, and the compatibility score. A **Close** button at the top brings the user back to the list. Nothing is changed in the data when this view is opened or closed. It must open within 200 milliseconds of the tap.

2.6 Refresh

The page shall have a **Refresh** button. When pressed, the system reloads the application list with the latest data from the database. This is useful because companies can change the status of an application at any time, and the user might not see the update unless they refresh. If loading fails, the page shall show the message "Unable to load your applications. Please try again." The list must update within 1 second of pressing Refresh.

2.7 Empty state

If the user has not applied to any jobs yet, instead of an empty list the page shall show the message "You haven't applied to any jobs yet. Head to the Recommendations page to get started." and a **Go to Recommendations** button that takes them there.

2.8 Loading state

While the data is being loaded, the page shall show a loading indicator instead of the list. The list must not appear until all the data is ready, to avoid showing incomplete information.

2.9 Navigate to Skill Gap Analysis

The Status Page shall have a **Skill Insights** button that takes the user to the Skill Gap Analysis page. The button must be visible without scrolling.

3. Skill Gap Analysis

3.1 Overview

The Skill Gap Analysis is a separate page that opens when the user presses the **Skill Insights** button on the Status Page. It looks only at the jobs the user was rejected from and checks what skills those jobs required compared to what the user has. The page must load within 1 second of opening.

3.2 Missing skills

The page shall show a list of skills that the user does not have at all but that appeared in the rejected jobs. The list is sorted by how many rejected jobs required each skill, with the most common ones at the top. Each entry shows the skill name and how many rejected jobs needed it (e.g. "Python - required in 5 rejected jobs").

3.3 Underscored skills

The page shall also show a list of skills the user does have, but where their score was lower than what the rejected jobs required. For each skill, the page computes how far off the user was on average across all rejected jobs that needed it, and sorts the list with the biggest gaps first. Each entry shows the skill name, the user's current score, and the average score that was needed (e.g. "SQL your score: 40 / average required: 75"). This section is labelled **Skills to Improve**.

3.4 Skill gap summary

At the top of the page, a short summary shall be shown with the total number of missing skills and the total number of skills to improve (e.g. "3 missing skills · 2 skills to improve"). If the user has no rejections yet, the page shall show: "No rejections yet keep applying to see your skill insights." If the user was rejected from some jobs but there are no skill gaps found, the page shall show: "Great news - your skills meet the requirements of all jobs you've applied to."

3.5 Refresh

The page shall have a Refresh button that reloads the data and recalculates the results. The page must update within 1 second of pressing it.

4) Company matchmaking page

The Matchmaking Page is the core of the application from a company's perspective. It presents applicants who have already expressed interest in one of the company's job postings, sorted by compatibility score, one at a time in a card-based layout. The company reviews each applicant and can accept or reject them with an action button.

Only applicants who have applied to one of the company's jobs appear here. The company does not see users who have not expressed interest. Applicants are served in descending order of compatibility so the strongest matches appear first.

When both the company and the user have expressed mutual interest (the user applied and the company accepted). If no applicants are available for review, an empty state screen is shown, prompting the company to refresh or check back later.

1. Card Stack

The matchmaking page displays applicant cards one at a time. The top card is always the next recommended applicant. Applicants the company has already reviewed are not shown again.

1.1 Display Applicant Card

When the company opens the matchmaking page, the next recommended applicant is loaded and displayed as the top card. The card must display:

- Applicant name
- Location
- Years of experience
- Education level
- The job title they applied for
- Compatibility score (as a percentage, 0–100%)
- Up to 5 of the applicant's skills with their score levels

The resume is not shown on the card preview. It is only visible in the expanded detail view. The card must be visible and interactive within 1 second of the page loading.

1.2 Accept: Advance Applicant

When the company presses the accept button, their interest in the applicant is recorded. The card animates off the right edge of the screen within 300 milliseconds and the next card becomes the top card.

1.3 Reject: Skip Applicant

When the company presses the reject button, the applicant is skipped and will not reappear. The card animates off the left edge within 300 milliseconds and the next card becomes the top card.

1.4 Action Buttons

Below the applicant card, two action buttons are displayed at all times:

- Red button → rejects the current applicant
- Green button → accepts the current applicant

The buttons are greyed out and inactive when no card is currently loaded.

1.5 Expand Card to Full Detail View

The company can tap anywhere on the visible card to expand it into a full detail view. The expanded view must open within 200 milliseconds of the tap. It shows all the information from the card preview plus:

- Full resume text
- All skills (not just the top 5) with their score levels
- Full job description of the job the applicant applied for
- Compatibility score breakdown
- Contact email
- Contact phone number

An Accept and a Reject button at the bottom of the view work the same as the main buttons. A back button closes the view and returns the company to the card stack without triggering any action.

1.6 Undo Last Action

The company can undo their most recent accept or reject by pressing the Undo button, shown as a back-arrow icon in the toolbar below the card. Pressing it reverses the last action and returns the card to the top of the stack. Only one undo is available per session. The Undo button appears greyed out and inactive at the start of a session and after an undo has already been used. The card must be restored within 500 milliseconds.

2. Empty State

2.1 Display Empty State Screen

When no more applicants are available for review, the card area is replaced by an empty state screen. The screen displays a descriptive message and a Refresh button that re-checks for new applications. The empty state screen must appear within 300 milliseconds of the last card being removed.

3. Validation

The accept and reject actions are only available when a card is currently loaded. Both buttons are greyed out and non-interactive when no card is present. Before processing an accept or reject, the system verifies that the application record still exists and is still in an applied state. If another company user has already processed the applicant from a different session, the company is informed and the next card is loaded automatically.

4. Privacy Unmasking

The candidate‘s Contact Email and Phone Number are hidden by default on the evaluation page. If the recruiter selects **Accept** and successfully submits, the privacy mask is removed and the contact information is revealed so the company can proceed to schedule an interview. If **Reject** is selected, the contact information remains permanently hidden.

5. Usability

All core actions (accepting, rejecting, and undoing) must be reachable within two taps from the matchmaking page.

6. Security

The matchmaking page only shows data for the currently authenticated company. All actions are validated against the active session before being executed. Companies must not be able to view or act on the applicants or records belonging to other companies through the interface.

5) Company Status Page

1. General Overview

The Company Status Page is the module through which a company reviews candidates who have been matched to their active job postings and have completed the testing phase. This page provides a structured hiring workflow: the recruiter can inspect each applicant's full profile alongside their technical test results, then issue an Accept or Reject decision, which is persisted to the Match table and triggers the appropriate privacy rules.

Test result data (candidate ID, test ID, and final percentage score) is consumed from the PowerPuff Girls' testing module, which validates and grades submissions. Once that data is available, this module integrates it into the applicant review view so the company can make an informed, evidence-based hiring decision.

2. Applicant List

The system shall allow companies to view a list of all current applicants who have matched with their active job postings. Each entry in the list displays the applicant's name alongside a **Review Applicant** button. Pressing this button navigates the recruiter to the individual applicant evaluation page described in Section 3.

3. Applicant Evaluation

When a recruiter presses **Review Applicant**, the system opens a dedicated evaluation page. All displayed data is fetched from the relevant modules and is read-only; it cannot be modified from this page. The following fields are presented:

Applicant Name, Location, Years of Experience, Education, Resume, Test Questions, Candidate Test Answers, and Automated Test Grade (an integer from 0 to 100 sourced from the PowerPuff Girls testing module).

After reviewing the information, the recruiter must fill in the two required decision fields before submitting.

3.1 Decision Fields

Decision is a required dropdown with the options **Accept** and **Reject**. **Feedback Message** is a required text field, limited to 500 characters, in which the recruiter writes the rationale for their decision. This message is sent to the candidate after submission.

4. Submitting a Decision

When the recruiter presses **SUBMIT** and all validation checks pass, the Match record status is updated in the database and the recruiter is navigated back to the applicant list. If any validation check fails, a pop-up informs the recruiter that at least one field is invalid, once dismissed, the evaluation page remains visible with all invalid fields highlighted by their red borders.

On a successful submission the system displays a confirmation pop-up with the message "Decision submitted successfully". If the submission fails due to an error, the pop-up instead shows "We're sorry, an error occurred. The decision was not saved. Please try again."

If the decision is submitted successfully, the date and time of the evaluation are automatically recorded as a timestamp field in the Match table.

5. Cancel Evaluation

The recruiter may cancel the evaluation at any time by pressing the **CANCEL** button at the bottom of the evaluation page. The system presents a confirmation pop-up asking "Are you sure you want to cancel the evaluation?". Pressing **Yes** returns the recruiter to the applicant list without saving any data to the database. Pressing **No** dismisses the pop-up and returns the recruiter to the evaluation page in its current state.

6. Data Integration with Testing Module

This module consumes the processed test result data produced by the PowerPuff Girls' testing module. Once a candidate completes and submits a technical assessment, the PowerPuff Girls' server validates the submission and computes the final percentage score. This module retrieves

the resulting record - consisting of the candidate ID, test ID, and final score - and surfaces it on the evaluation page alongside the candidate's own written answers, so the recruiter has both the automated grade and the raw answers available for review.

6) Cooldown System Requirements

Recommendation prevention

The system shall prevent a rejected job recommendation from reappearing in a user's recommendation queue for a configurable cooldown period (default: 24 hours) after it was last shown.

Record shown

The system records the exact DateTime at which a job was displayed to a user.

Check cooldown

When the system looks for the next recommendation to display, it checks whether the user was shown that recommendation before. If yes, and not enough time has passed, the system skips it.

7) Chat System Requirements

The chat system enables real-time messaging between users and companies, and between users and other users, within the Tinder Jobs platform. Employers can initiate conversations with any candidate. Users can message companies directly or message other users directly.

1. Chat Access & Initiation

Employers can initiate a conversation with any candidate on the platform without requiring a prior match or application. Users (job seekers) can message any company directly or message other users directly. When a user initiates a conversation with a company, they may optionally select a job listing to associate with the conversation. If a job is selected, it is displayed prominently within the chat thread so both parties have clear context about the role being discussed. If no job is selected, the job banner is hidden and the conversation proceeds as a general enquiry. Employers initiating a conversation are not required to associate a job listing.

2. Chat Inbox & Organization

The chat inbox is divided into two separate tabs: Users and Company. The tabs are client-side filters on the same inbox - they do not change the logged-in user's identity or active mode. The Users tab displays conversations between the current user and other individual users. The Company tab displays conversations between the current user and companies. Each tab lists conversations sorted by recency, with the most recently active thread appearing first. A conversation moves to the top of the list only when a new message is sent or received - opening a chat without sending a message does not change its position.

Each conversation entry displays the other party's name and avatar, the last message snippet, and the timestamp of the last message.

When the logged-in user is acting in Company mode, the inbox displays a flat list of all conversations the company has with users. No tab switcher is shown in Company mode since companies can only converse with users.

2.1 Search

A search bar is available above the conversation list. The search is scoped to the active tab - searching in the Users tab queries other users, searching in the Company tab queries companies. In Company mode, the search always queries users. Results include both existing conversations matching the query and new contacts not yet in the inbox. Existing conversations appear at the top of the results dropdown. Selecting an existing conversation opens it without changing its position in

the list. Selecting a new contact creates a new chat thread and inserts it at the top of the list. In User mode, the inbox automatically switches to the correct tab after a new chat is created.

3. Supported Message Types

Users and companies can send text messages, images, and file attachments within a conversation. File attachments are intended primarily for CVs and documents relevant to a job application. All attachment types must be validated before sending to prevent unsupported formats from being uploaded.

3.1 Text Messages

Plain text messages can be composed and sent by any participant in a conversation. Messages are limited to 2,000 characters. No Markdown or rich text formatting is supported.

3.2 Images

Users may attach and send images within a chat. Supported formats are JPEG and PNG. Images must not exceed 10 MB per file. Images are rendered inline within the message thread upon receipt.

3.3 File / CV Attachments

Users and employers may send file attachments, including CVs and other relevant documents. Supported file formats include PDF, DOC, and DOCX. Files must not exceed 20 MB per attachment. Attachments are displayed as a downloadable link within the message thread.

3.4 Attachment Selection

Attachments are selected via the operating system's native file manager dialog (Windows OpenFileDialog). The dialog is pre-filtered to supported formats only: JPEG, PNG, PDF, DOC, and DOCX. No intermediate attachment type picker is shown to the user. The message type (Image or File) is determined automatically from the selected file's extension.

4. Read Receipts

Each message displays a timestamp indicating when it was sent. In addition, the sender's most recent sent message displays a delivery and read status indicator beneath it. Once a message has been delivered to the recipient's device, it is marked as Delivered. Once the recipient opens and views the message thread, all unread messages in that thread are marked as Seen and the

indicator updates accordingly. Only the last sent message shows the Delivered or Seen label - all other messages display their timestamp only.

5. Blocking & Chat Management

Users can block another user or company from within a conversation. Blocking prevents the blocked party from sending further messages and removes the conversation from the blocked party's inbox. The blocking user retains the chat history in their own inbox unless they also choose to delete it. Users can permanently delete a conversation from their chat history, removing it from their inbox view. Deletion is local to the user who performs the action and does not affect the other party's view of the conversation.

5.1 Blocking

A block can be applied to any individual user or company from within the conversation's dropdown menu. Once blocked, the blocked party cannot initiate new conversations or send messages to the user who performed the block. If the blocked party attempts to send a message, they will not be notified that they have been blocked. The block can be removed at any time, but only by the user who originally applied it. Blocking does not automatically delete the conversation.

5.2 Chat Deletion

Users can delete a conversation from their own inbox permanently. This action is irreversible and removes all messages from that thread in the deleting user's view. A confirmation dialog is presented before deletion to prevent accidental removal.

6. Chat Header & Conversation Actions

Each open conversation displays a header showing the other party's avatar and name. If a job listing is linked to the conversation, a job banner is displayed prominently below the name. If no job is linked, the banner is hidden entirely.

A dropdown menu button is located in the top-right corner of the chat header. Clicking it opens a menu with the following actions, shown conditionally based on context:

- Block - visible when the conversation is not currently blocked. Blocks the other party.
- Unblock - visible when the conversation is blocked and the current user is the one who applied the block. Removes the block.
- Delete - always visible when a conversation is open. Permanently removes the conversation from the current user's inbox after confirmation.

- Go to Profile - visible when the other party in the conversation is a user. Navigates to that user's profile page.
- Go to Company Profile - visible when the other party in the conversation is a company. Navigates to that company's profile page.
- Go to Job Post - visible only when a job listing is linked to the conversation. Navigates to the relevant job post page.